Theory of Computation • 1-7 • Review of Mathematical Theory

Let us discuss cach one by one

i) Negation

This prnoposition is also called NOr proposition. It is denoted by 1 or = or having horizontal bar on letter. If S is a proposition then IS defines NOT S. The truth table for this connective is

S	1s
T	F
F	T

i) Conjunction (AND)

If A and B are two propositions then conjunction of A and B is denoted by A^B.The truth table for this connective is

A	B	A^B
T	T	T
T	F	F
F	T	F
F	F	F

i) Disjunction OR

If A and B are two propositions then disjunction of A and B is denoted by AvB.The truth table for this connective is

A	B	AvB
T	T	T
T	F	1
F	T	T
F	F	F

iv) Implication

If A and B are two propositions then if A then B is a proposition.It can be denoted byA-BorA = B.

1-8 • Review el Mathematical Theory

Theory of Computation

r this conco			
			A
A			
		T	T
T			
		F	F
T			
F		T	T
F		F	T

For exampleFind trth value for following proposition f an apple is not green then it must be red. Let A denotes An apple is not green and B denotes It must be redThe statement A is false and B is true. Hence the proposition is true

v) If and only if

f A and B are two propositions then A if and only if B.I can be denoted as AR The truth table will be

A	B	AB
T	T	T
T	F	F
F	T	F
F	F	T

For example: I will go to the school if and only if it is not rainingHere statement A is I will go to the school. The statement B is It is not rainingThus the resultant proposition is false.

1.5 Relations

Relaonship is a major aspect between two objectseven this is true in our real lif One ohect can be related with the other object by a some relation.Then those two objes forma pair based on this certain relationship. DefinitionThe relation R is a collction for the set S which repreents the pair of elements.

pair is chosen from a set called range

Theory of Computation • 1-9 • Review of Mathematical Theory

Properties of Relations

A relation R on set S is

1. Reflexive if iRi for all i in S.
2. Irreflexive if iRi is false for all i in S.
3. Transitive if iRj and jRk imply iRk

.Symmetric if iRj implies jRi.

5. Asymmetric if iRj implies that jRi is false.

Every asymmetric relation must be irreflexive.

For example: if A = ab |then

reflexive relation R can be = a, a.b, b1
irreflexive relation R can be = ab1
transitive relation R can be = a, bb.a).aa
symmetric relation R can be = a.b).b, a
asymmetric relation R can be=|a, b1

Equivalent Relation

A relation is said to be equivalence relation if it is reflexive, symmetric and transitive,

over some set S.

Suppose R is a set of relations and S is the set of elements.

For example: S is the set of lines in a plane and R is the relation of lines intersecting

to each other.

A relation is said to be antisymmetric if a, be R and b, ae R then a =a

Example

1.5: Determine whether R is equivalence relation or not where

A = 0121R=/001, 011)2,

2)2, 1

Solution: The R is reflexive because 0,

0)1,

1)2,

2) e R.

Where as R is not symmetric because (0, 1 is not in R whereas 1,

0) is in R

Hence the R is not a equivalence relation.

Closures of Relations

Sometimes when the relation R is given it may not be reflexive or transitive

By adding some pairs we make the relation as reflexive or transitive.

For exampleLetab)bca, abb